SVM Regression and SONN based approach for seasonal crop price prediction

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Abstract

In Agriculture pre harvest glut, post harvest loss and intermediary involvement cause producer to get the lowest price in the entire marketing process. Though the authorities provide long time monthly price series for various geographically spread regions of the country in websites, the lack of knowledge to process online market data does not allow farmers to take a dynamic decision for the sale of their produce. In this model, the equipped marketing information is analyzed for integrating domestic markets and manifesting price transmission from markets to farm gate (prices received by farmers) leading to sustainable profit. Market wise support price for crops is extracted from website tables using XML SAX(Simple API for XML) Expression and interested crop price associated with its season and variety are clustered using SONN (Self organized neural network). Then support vector regression (SVR) based price prediction for long and midterm movement helps farmers in the decision making of agricultural marketing. As all the processing is done at the back end it is easy for the farmers to interact with the system and reach for more accurate prediction. The results assure of more precision than other traditional methods such as regression techniques and moving averages

Keywords: *XML SAX Expression, price extraction, SONN, clustering, SVR, prediction, sustainable profit*

1. Introduction

Indian Agricultural Market structure is a network of cooperatives at the local, regional, state and national levels. Over a period of time, these markets have acquired the status of limiting and monopolistic, providing no help in direct and free marketing, thereby preventing private investments in the sector. The misuse by intermediaries, exporters, processors and traders prevents farmers from getting premium price and timely payment for their produce **[4]**. Nowadays, finding a suitable market for the marketed surplus with the increased agricultural production is a big challenge. For example, if crop production during the season is surplus, it ends up in post harvest loss of product. But if the same product is transported to another market where there is a need and preference, considerable profit could be achieved. The changing global agricultural scenario insists on review of the policies related to pricing, marketing and trading of agricultural commodities. In order to provide dynamism and efficiency in the marketing system online market wise monthly series price data were made available for direct usage by farmers. This amplified marketing information has a positive effect in decision making, but collecting and disseminating information has been often complicated and expensive. Data mining techniques have a great deal of interest for price forecasting in recent years that aids huge amount of data processing for better price realization [3, 8].

Many studies have been undertaken for financial prediction using various regression techniques and rule based systems. In data mining perspective, yield prediction can be formulated and treated as a multi-dimensional regression task [8].Technical aspects, requirements of common mining tools are discussed and data mining methods related to data management, analysis techniques are compared with each other [1].In Credit card payments more insight into performance abilities are evaluated for different data mining techniques [3].The quadratic forecasting model of linear time series compares the prediction results of different time series and different sample data to find the best forecasting model to forecast the price in cites [2]. Regression techniques such as the Ordinary Least Squares (OLS), Principal Component Regression (PCR) and Latent Root Regression (LRR) methods are mentioned and their relevance due to shelf life prediction is discussed [5]. The summer daily rainfall model shows the feasibility of the BART ensemble model in improving the accuracy of torrential rainfall with linear regression and nonlinear regression model [6].For effective prediction of heart attack, Multi-layer perceptron neural network with selected significant patterns uses Back-propagation as the training algorithm [9]. Dealing with uncertainty in finance primarily involves recognition of patterns in data and using these patterns as future events for prediction ranks as the most critical criteria for financial survival [10].Textual articles that appear in the leading and influential financial newspapers are taken as input. From those articles the daily closing values of major stock market indices in Asia, Europe and America are predicted, thereby exploiting textual information in addition to numeric time series data [11] . The potential of coactive neuro-fuzzy inference system (CANFIS) for estimation of flow through trapezoidal and rectangular rockfill dams and illustrates accurate flow predictions can be achieved with a CANFIS with the Takagi–Sugeno–Kang (TSK) fuzzy model and the Bell membership function for both trapezoidal and rectangular rockfill dams .In SVR the input training data are nonlinearly mapped into a high dimensional space [15] .The genetic algorithm is used to optimize the parameters of SVR and its ability is controlled by its parameters. The wear rate data of 20 mileages are employed to study the lifetime prediction of the cylinder by the GSVR model. In order to prove the superiority of GSVR in lifetime prediction of the cylinder, the RBF neural network and BP neural network are employed to compare with GSVR. The results of the experiments show that the lifetime prediction model of the cylinder by the GSVR is better than that of the RBF neural network and BP neural network [16].

Most of the financial decision making applications use static history of data for statistical predictions. Automatic learning systems infer rules from appearance and behavior of data entities at run time. These rule based soft computing and regression technologies concentrate on static data which are less successful .Self organized neural network (SONN) when used in clustering is typically a collection of neuron-like processing units with weighted connections between the units. Due to its dynamic training attributes, association and alteration acts as an alternative approach for financial predictions that are difficult to be solved by means of conventional approaches [8]. Support Vector Machines (SVM), a form of supervised learning predicts new values from a series of old data. On the whole, the proposed dynamic crop price extractor using SONN increases the accuracy of precipitation in forecasting agricultural wholesale or retail market prices

2. Architecture

For price prediction, accuracy and robustness is essential to reduce the error between actual and forecasted data. The selling prices of crops (target data) are published on the agricultural product marketing websites every day. Our model integrates XML DOM with SONN, Support vector regression to filter interested price data and forecast market prices. SONN

divides the set of existing market price data into interested groups based on season of cultivation and its varieties. SVR algorithm learns the relationship between the observed crop price and the underlying market conditions thereby discovers the pattern for crop trading relative to the market. The block diagram of the proposed system is given in Figure 1. It has the following four phases.

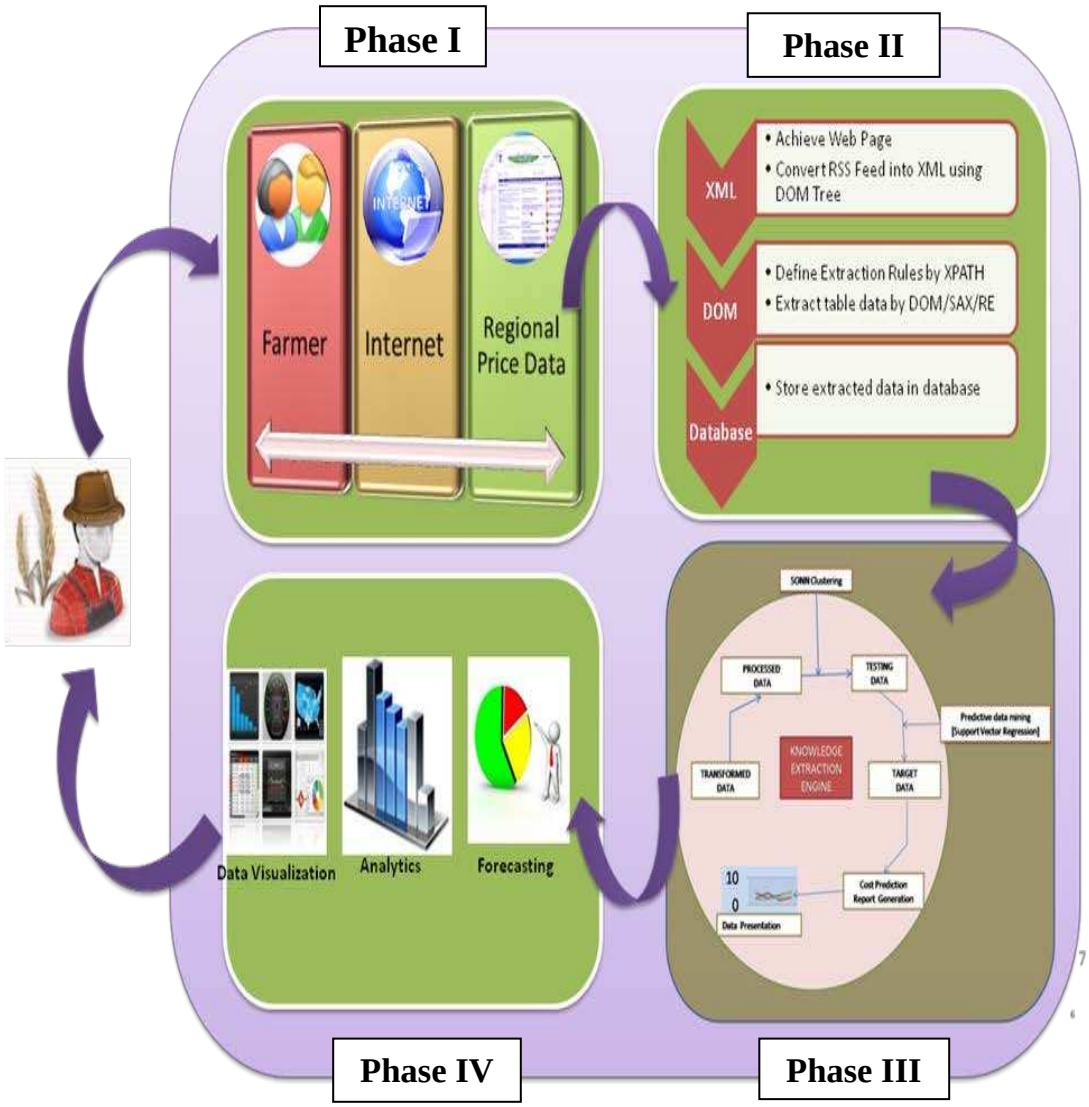


Figure 1. Architectural Model for Dynamic Prediction

2.1. Phase-I

The user opens the URL containing online market price.

2.2. Phase II

It detects the RSS Feed on web pages and transforms it into a DOM tree, Further stores the node values of the "<td>" tags by defining extraction rules .The stored data is given as input to the knowledge extraction engine for preprocessing and prediction.

2.3. Phase III

Marketing analysts may wish to categorize a group of agricultural commodities to buy, sell, or hold. Neural network technology is widely used to cluster such complex and numerous unrelated variables. In Knowledge extraction engine, the transformed data are preprocessed to deal with null values and outliers. SONN are interconnected networks of sovereign processors which dynamically alter training attributes and their associations to discover data in clustered pattern. The support vector regression technique runs on the testing data to calculate the regression coefficient values. It also calculates the predicted price data and stores them along with the regression coefficient values. This process operates in parallel with the data storage to economize number of database accesses**.**

2.4. Phase IV

The resultant data can be utilized by farmers for the following decision makings:

1. Crop cultivation
2. Visualization of future price
3. Selection of market for crop marketing

3. Functional Model

3.1. Empirical Data

Real–time crop marketing data has been obtained from Agritech portal of Tamilnadu Agricultural University from 2003 to 2011 by observing 32 districts, with 1025 samples are clustered as training data for Vellore District (Figure 2), other samples are tested for 2012 using Support Vector Regression.

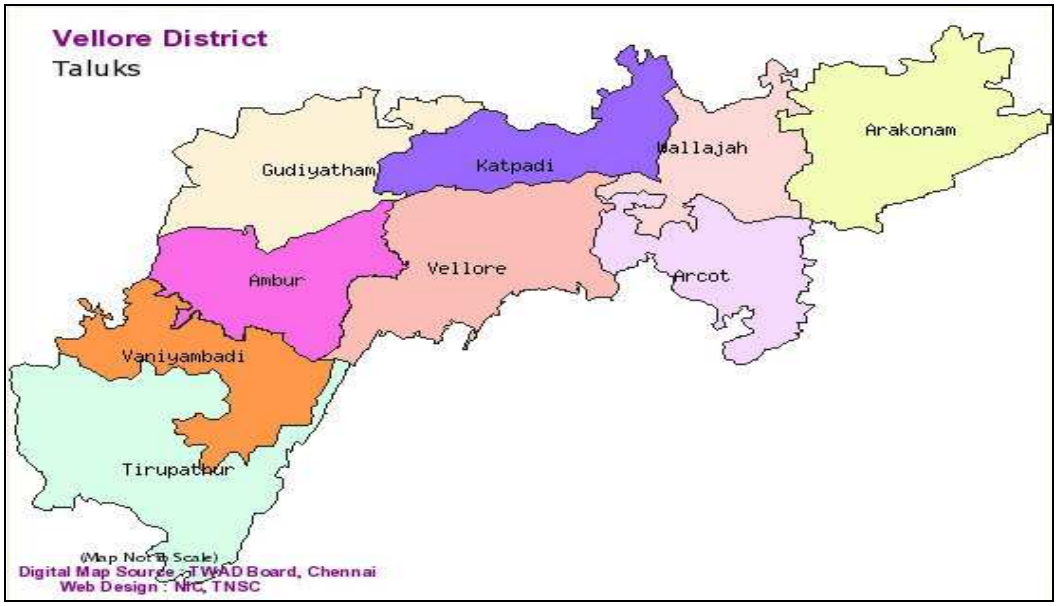


Figure 2. List of Commodity Markets in Vellore District [courtesy:http:// www.twadboard.gov.in/twad/vellore_dist.aspx]

3.2. Extraction of Dynamic Marketing Information from Web Pages using DOM

Many websites organizes publicly accessible contents as RSS Feed (Really Simple Syndication) for interested persons. RSS publishes frequently updated contents (*i.e.*, price information) in standardized format. The following algorithm filters the featured price from web pages/RSS using DOM (Domain object model)

INPUT: Agritech Portal URL,

OUTPUT: Filtered Featured Price data

Reach the RSS tags of a Web page. Ignore the rest

FOR i=< Root > to </Root> tag|

Obj=Generate DOM tree

END FOR

FOR i=<table> to </table> tag

IF Node=”< TR>” THEN

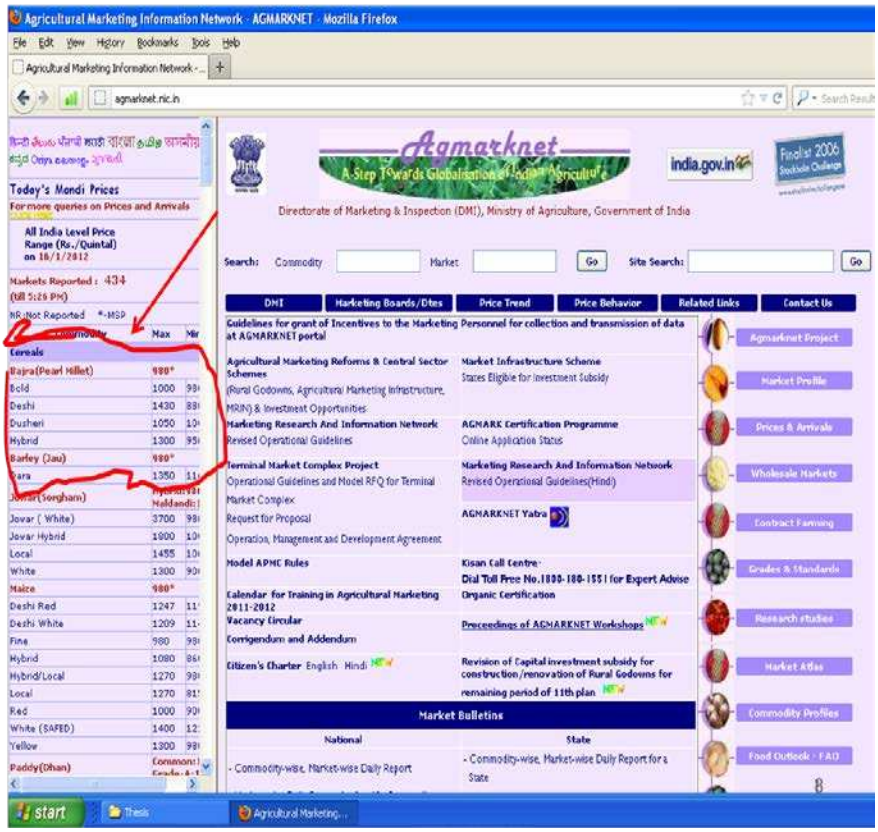
NEXT NODE

ELSEIF Node=”<TD>” THEN

```
        xmlDoc.documentElement.childNodes
       Obj1=xmlDoc.documentElement.childNodes;
for (i=0;i<x.length;i++)
 {

    document.write (“  ”);
    document.write(x[i].childNodes[0].nodeValue);
    document.write("<br>");
    Insert node Value in Excel
  }
```

END FOR



Market	Date	Variety	Modal price	Max price	Min price
Ammoor	1/2/2003	ADT 37	553	747	650
Nagapatinam	2/4/2003	ADT 36	639	671	655
Ammoor	2/5/2004	ADT 36	639	671	655
Kaveripakkam	4/6/2005	ADT 36	675	672	672
Namakkal	4/8/2006	BPT	1015	1105.54	978
Thanjavur	4/7/2007	Other	915.03	882.46	810
Cuddalore	3/9/2008	A. Ponni	625	684	480

Figure 3. Filtered Sample Training Data from Online RSS Feed [Data courtesy http://www.agmarket.nic.in]

3.3. SONN Clustering for Commodity Categorization

Clustering is a process of unsupervised learning which aims at discovering a new set of physical or abstract objects into reference set of classes. The cluster has the characteristics of similarity within the group and is described as unsupervised learning of a concealed data

concept. Let the set of *n* points $\{x_1, x_2, \ldots, x_n\}$ be represented by the set *S* and the K clusters be represented by $C_1, C_2, \ldots, C_K$. Then

$$C_i \neq \phi \qquad for\ i = 1,.....,k,$$

$$C_i \cap C_j = \phi \qquad for\ i = 1,.....k,\ j = 1,.....,k \ \ and\ i \neq j \ \ and \bigcup_{i=1}^{k} c_i = s$$

In SONN clustering [17] an input pattern with n features is represented by a vector ***x*** in a *n*-dimensional pattern space. The network maps the input pattern to an output space. The output space is a 1-dimensional or 2-dimensional array of output nodes. The neurons are organized in a lattice (one or two-dimensional) is placed in the input space and is spanned over the input distribution. Each processing unit in the SONN lattice is associated with a weight vector of dimension similar to input vectors.

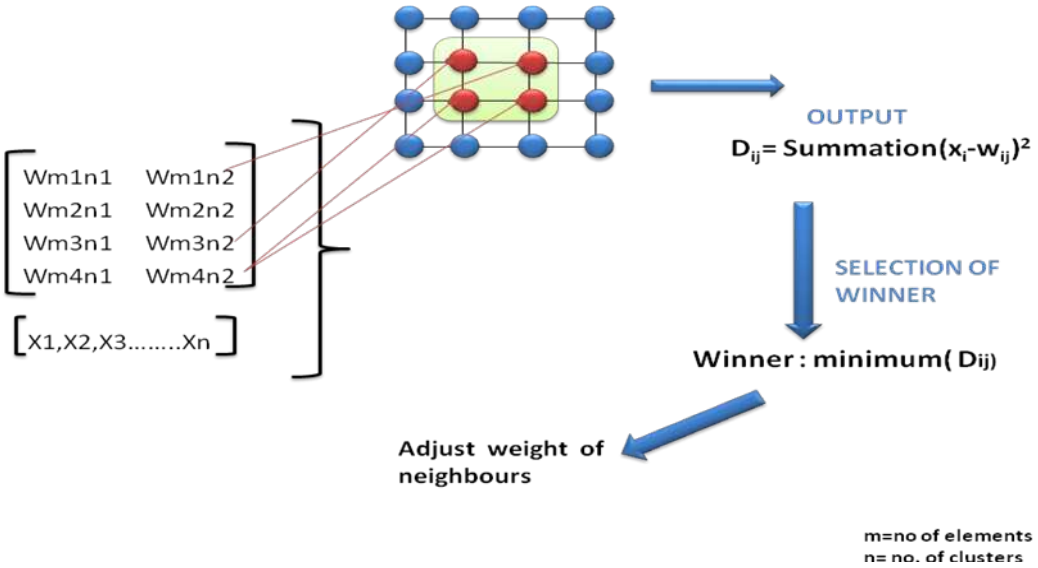


Figure 3. SONN based Clustering

Using the weights of each processing unit as a set of coordinates the lattice can be positioned in the input space. During the learning stage the minimum Euclidian distance between weight vector and the input vector 'x' shows that input vector 'x' is close to corresponding output vector and is considered as the winner .Accordingly the weights of the units change their position move towards the input points. When the map is visualized the inputs can be associated with each cell on the map. One or more cell that clearly contains similar documents can be considered as a cluster on the map. These clusters are generated during the learning phase without any other information.

Here SONN divides the set of *N* existing market price data into *M* interested groups based on season of cultivation and its varieties so that the target data size and variability is minimized (Figure 4).

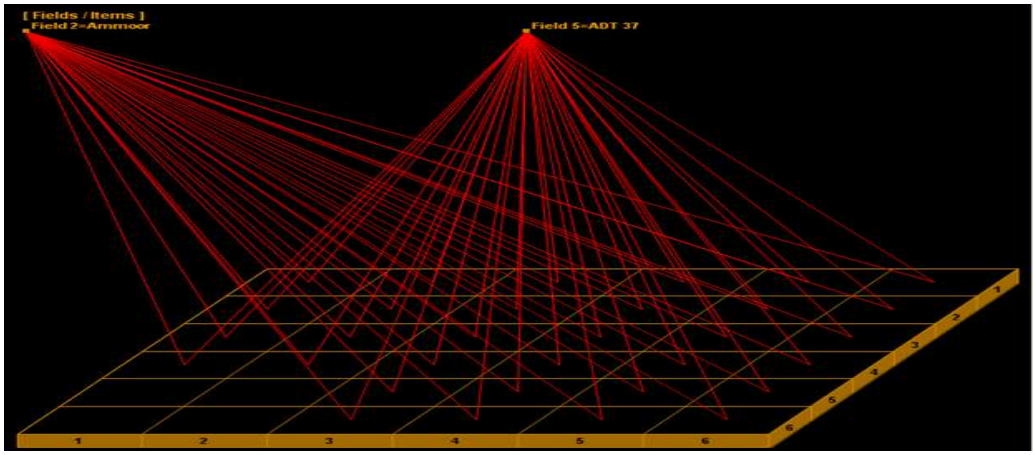
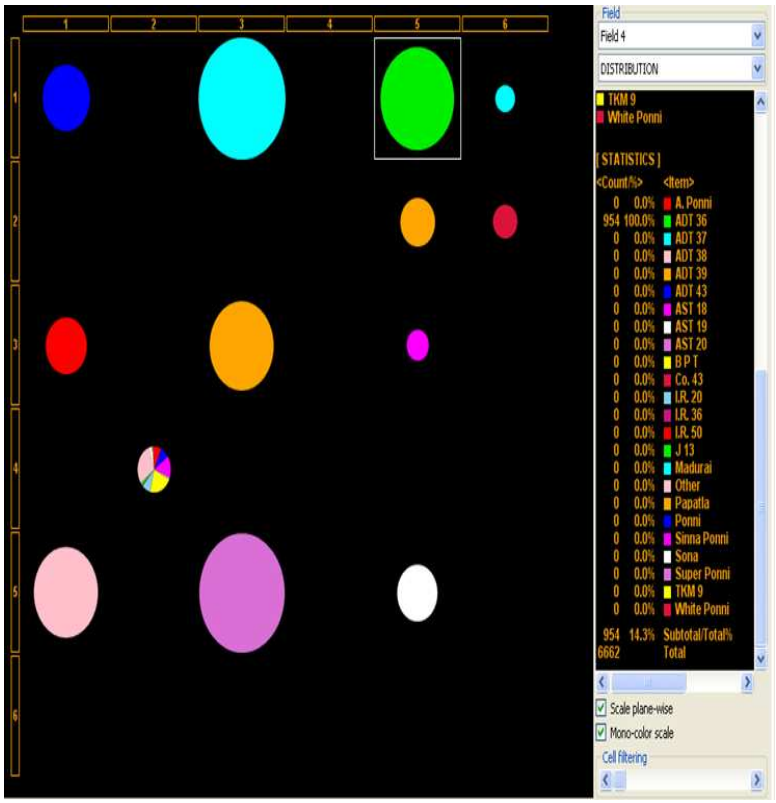


Figure 4a. Distribution Graph for Ammoor Market



Variety	Distribution Count
A.Ponni	1280
ADT 36	954
ADT 37	718
ADT 38	726
ADT 39	398
ADT 43	311
ADT 18	298
ADT 19	291
ADT 20	219
Co.Ponni	191
IR 20	106
Sinna Ponni	86
White Ponni	7

Figure 4b. Clustered Variety Wise Price Data for Ammoor Market

3.4. Support Vector Regression based Crop Price Prediction

SVM Regression (SVR) is a method to estimate a function that maps from an input object to a real number based on training data [15]. A training set for regression is represented as follows

$$D = \{(\mathbf{x}1, y1), (\mathbf{x}2, y2), \ldots, (\mathbf{x}m, ym)\} \quad (1)$$

where $\mathbf{x}_i$ is a *n*-dimensional real vector, y_i is either 1 or -1 denoting the class to which the point $\mathbf{x}_i$ belongs. SVR function $F(\mathbf{x})$ approximates all pairs ($\mathbf{x}_i$, y_i) and maintains the differences between estimated values and real values under e precision. That is, for every input vector $\mathbf{x}$ in *D*,

$$F(\mathbf{x}) = \mathbf{w}\cdot\mathbf{x} - b \tag{2}$$

subject to

$$y_i - \mathbf{w}\cdot\mathbf{x_i} - b \le \mathrm{e} \text{ and } \mathbf{w}\cdot\mathbf{x_i} + b - y_i \le \mathrm{e} \tag{3}$$

The distance from the hyper plane to a vector $\mathbf{x}i$ is formulated as

$$\text{Margin} = \frac{1}{\|w\|} \tag{4}$$

By minimizing $\|w\|^2$ to maximize the margin, the training in SVR becomes a constrained optimization problem as follows.

$$minimize: L(w) = \frac{1}{2}\|w\|^2 \tag{5}$$

$$subject\ to: y_i - w.x_i - b \le e \tag{6}$$

$$w.x_i + b - y_i \le e \tag{7}$$

To allow some errors to deal with noise in the training data, The soft margin SVR uses slack variables $\pounds_i$ and $\hat{\pounds}_i$. Then, the optimization problem can be revised as follows.

$$minimize: L(w, \pounds) = \frac{1}{2}\|w\|^2 + C\sum_i\left(\pounds_i^2\,\hat{\pounds}_i^2\right),\ C > 0 \tag{8}$$

$$Subject\ to: y_i - w.x_i - b \le e + \pounds_i,\ \forall(x_i, y_i) \in D \tag{9}$$

$$subject\ to:\ w.x_i + b - y_i \le e + \pounds_i,\ \forall(x_i, y_i) \in D \tag{10}$$

$$\pounds, \pounds_i \ge 0 \tag{11}$$

The constant $C > 0$ is the trade-off parameter between the margin size and the amount of errors. Finally, the SVR function $F(\mathbf{x})$ becomes the following using the kernel function.

$$F(x) = \sum_i (\propto_i - \propto_i) K(x_i, x) + b$$

In our experiment three input variables including Crop variety, Season of cultivation and market price are deemed as independent variables and crop price fluctuation is deliberated as the dependent variable. The parameter values used here are C = 60 (tradeoff between training error and margin), standard deviation C= 4.0 and slack variable C= 0.2

4. Experimental Results

The Dynamic Agricultural marketing data set is preprocessed successfully by removing duplicate records and supplying missing values. The refined data set, resultant from preprocessing, is then clustered variety wise for vellore district of tamilnadu using SOM clustering via CMSR-Cramer Modeling Segmentation and Rules software. For SVM, training

data include the seasonally separated price data from 2003-2011, is tested for the year 2012 and comparative results are introduced in table I. In Figure 5 the outcome of the model and its results are graphically represented for year 2012. In order to measure the accuracy of forecast by time series data pattern, the error (E) is used. If Y_t is an indicator of the real value of variable in time (t) and $Y_{t^{\wedge}}$ is an indicator of the forecasted value of the variable, n is the number of population then the error can be formulated as $E= Y_t - Y_{t^{\wedge}}$

Ammor (Rs /Quintal) Paddy(ADT 37)	
Actual Min/Max	Projected Min/Max
890/1136	948/1172
934/1080	925/1102
890/1035	863/1050
767/975	754/962
881/1091	846/1058
969/1363	978/1399
1155/1568	1122/1577
1117/1563	1132/1156
1228/1615	1250/1604
1323/1671	1359/1729
1300/1595	1319/1560
890/1136	909/1145

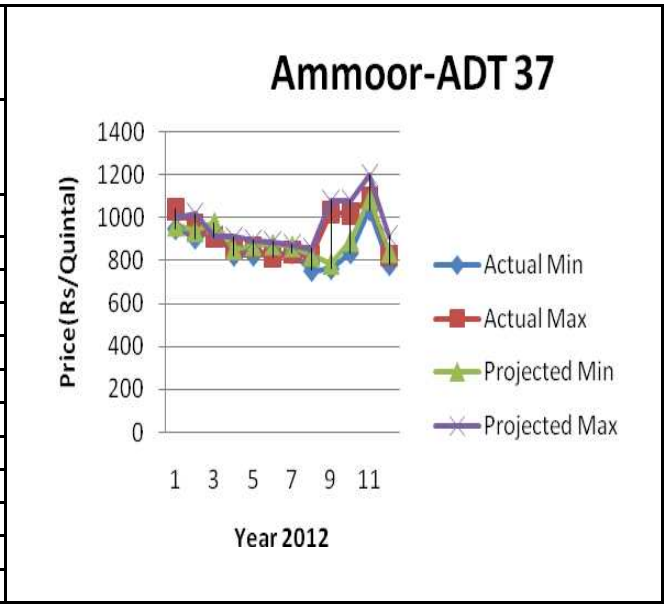


Figure 5a. SVM based Predicted Results for Ammoor Market

Kalavai (Rs /Quintal) Paddy(ADT 37)	
Actual Min/Max	Projected Min/Max
932/979	899/943
907/1004	923/982
872/943	894/928
990/1142	1026/1175
1029/1313	1038/1304
1318/1475	1338/1510
1292/1475	1257/1488
1286/1475	1273/1502
1286/1430	1259/1439
1386/1466	1377/1408
1449/1562	1507/1567
932/979	941/992

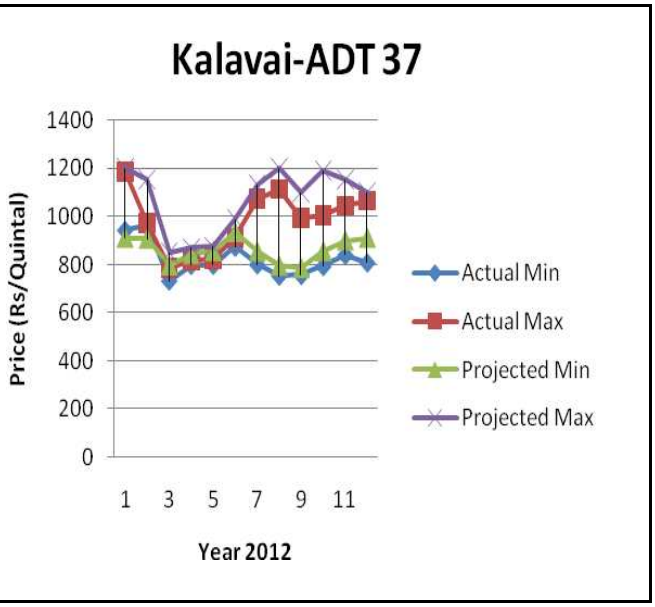


Figure 5b. SVM based Predicted Results for Kalavai Market

Katpadi (Rs /Quintal) Paddy(ADT 37)	
Actual Min/Max	Projected Min/Max
714/864	772/894
772/941	712/954
837/916	828/937
724/842	697/881
769/863	712/902
928/1130	910/1108
1074/1319	1042/1400
1014/1201	975/1276
1237/1264	1223/1291
1339/1441	1318/1450
1208/1346	1221/1399
714/864	744/806

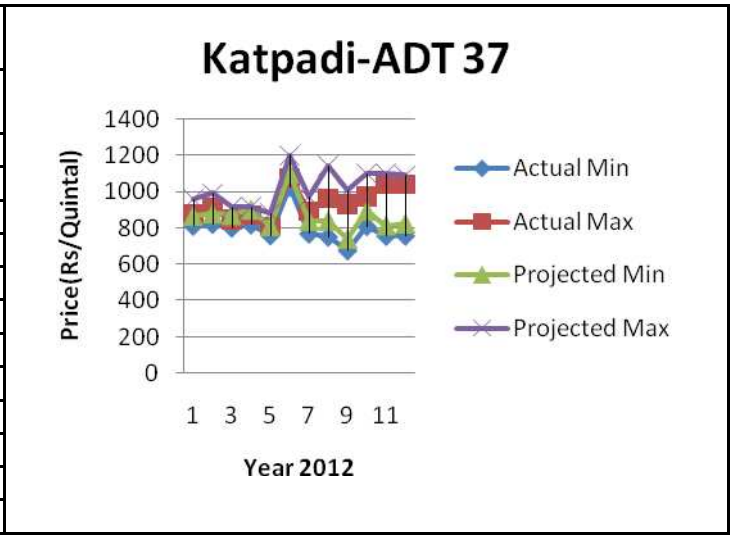


Figure 5c. SVM based Predicted Results for Katpadi Market

Kaveripakkam (Rs /Quintal) Paddy(ADT 37)	
Actual Min/Max	Projected Min/Max
913/937	926/959
930/1011	900/1050
887/948	908/1007
745/860	664/869
887/972	860/914
1035/1172	978/1183
1194/1362	1185/1401
1155/1436	1213/1478
1310/1419	1290/1437
1376/1541	1337/1511
1279/1413	1257/1426
913/937	884/987

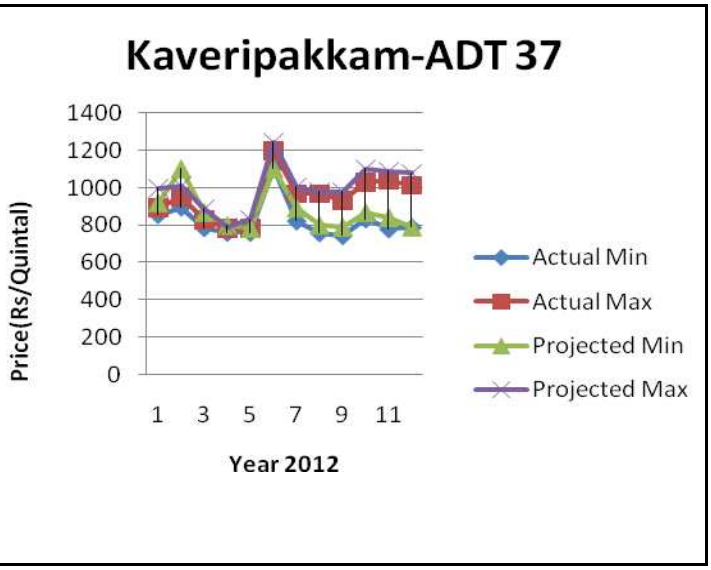


Figure 5d. SVM based Predicted Results for Kaveripakkam Market

4.1. Performance Measurement

From Figure 5, we visualize that prediction accuracy of Support Vector Regression model is much higher. The comparative results (Table 1) shows that SVR outperforms the other two models under the same input. Furthermore, SVR takes the least amount of computation time and performs best on most of the data sets, regarding of error measures.

RMSE (Root mean squared error) = $\sqrt{\sum_{i=1}^{n}(Yt - Yt^{\wedge})^2}/n$

MSE (Mean squared error) = $\sum_{i=1}^{n}(Yt - Yt^{\wedge})^2/n$

MAE (Mean absolute error) = $\sum_{i=1}^{n}(Yt - Yt^{\wedge})/n$

MAPE (Mean absolute percentage error) = $\sum_{i=1}^{n}(Yt - Yt^{\wedge})/n) * 100$

Table 1. Comparative Analysis of Performance Measures

Support vector Regression		**RBFF**		**Neural Networks**	
RMSE	14.10	RMSE	62.09	RMSE	15.43
MSE	208.14	MSE	3855.4	MSE	228.33
MAE	9.06	MAE	50.37	MAE	19.23
MAPE	1.16	MAPE	6.36	MAPE	4.17

5. Conclusion

With the advancement of technology in machine learning, it is impossible for farmers to get the algorithms out of the laboratory and put into the working environment for obtaining benefits. Here an approach is designed for them so that their simplified input (interested crop, season, market) to the model discovers knowledge of crop price specific to their environment. This driven approach includes current neural network techniques to increase profits, awareness, performance and stability at minimal cost. The results also illustrate that it has better precision than conventional neural network methods.

6. Future Enhancements

In future, the proposed research design can be enhanced by incorporating stock availability and its demand to unleash the hidden patterns thereby forecasting its future prices. Features such as platform compatibility in mobiles and tablets, personalized crop prices prediction, advertisements at farmer gate, online conversation with experts *etc.*, enables all analysis based deliverables as exclusive farmer views at minimal cost.

Acknowledgements

The authors would like to thank the Editor in Chief, the Editorial Secretary and the Reviewers for making this work much better than earlier. They also express their regards to Doctoral Committee members for the extended support.

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